

Prior Calibration Confounds Dose-Response Evaluation of LLM Forecasters

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Abstract

Dose-response evaluation, in which graded amounts of evidence are shown to a forecaster to see how its probability moves, is a natural test of whether an LLM forecaster reasons from new information. On 40 resolved binary questions across 7 frontier LLMs (3,360 elicitations), we show that this evaluation does not work in its natural form: per-model update slope is mechanically bounded by prior calibration (Proposition 3.1), so slope rankings cannot be read as skill rankings without prior control. Against a Manifold human-crowd anchor on the same questions, LLMs close only 39.6% of the gap between their own prior and the late-trading human consensus (CI [0.256, 0.490], 239 pairs). Independently, a crossed random-effects decomposition attributes 85.2% of per-(model, question) Brier@L3 variance to question identity versus 0.5% to model identity (175:1). We introduce DRR^{Res} , the prior-controlled per-model resolution gain, as the right axis on which to compare LLM updating skill; honest dose-response evaluation requires prior control, human anchors, and question stratification.

1. Introduction

The recent literature on AI forecasting has converged on two empirical claims: (i) ensembles of frontier LLMs match the human crowd on contemporary benchmarks (Halawi et al., 2024; Schoenegger et al., 2024), with some agentic systems reaching superforecaster parity (Alur et al., 2025), though expert forecasters still outperform top LLMs on rigorous benchmarks (Karger et al., 2024); and (ii) the evaluation methodology underpinning these claims is fragile, with simulated-ignorance prompts (Li et al., 2026), retrieval

contamination, and benchmark gaming (Paleka et al., 2025) all undermining post-hoc Brier comparisons.

We propose a third axis: rather than asking how well an LLM *predicts*, ask how its forecast *changes* when given graded evidence. We operationalise this with a four-level *evidence ladder* per question and elicit forecasts from a panel of 7 reasoning and non-reasoning frontier LLMs. This paper makes one argument with three supports: **the natural form of dose-response evaluation does not work; a properly-anchored version produces a sharp negative result on LLM updating skill; and after both fixes, model identity explains a tiny fraction of the remaining variance.**

Support 1 (structural; Proposition 3.1): Per-question slope $\beta_{m,q}$ is mechanically bounded by a function of the prior calibration $a_{m,q,0}$: under monotone ladder trajectories, $\beta_{m,q} \leq K_L(1 - a_{m,q,0})$ with K_L depending only on the number of rungs (Proposition 3.1). Since truth-aligned $\text{Brier@L0} = (1 - a_{m,q,0})^2$, the bound is equivalently $\beta_{m,q} \leq K_L \sqrt{\text{Brier@L0}}$, forcing a positive coupling between slope and prior miscalibration regardless of model skill. Empirically the OLS coefficient of β_m on $\sqrt{\text{Brier@L0}}$ is 0.131, consistent with the bound. Naive slope-skill correlations are therefore unidentifiable at the cross-model panel sizes typical of forecasting evaluations.

Support 2 (Manifold anchor): Both the total-vs-total ratio $(a_{m,q,3} - a_{m,q,0})/(P_{\text{late}} - P_{\text{early}})$ and the marginal-from-prior ratio $(a_{m,q,3} - a_{m,q,0})/(P_{\text{late}} - a_{m,q,0})$ yield a median capture of $\sim 39.6\%$, with bootstrap 95% CIs bounded well away from 1.0. The agreement between metrics addresses the natural concern that the total ratio conflates LLM prior-knowledge with conservative updating: removing prior-knowledge from the denominator does not change the headline.

Support 3 (variance bottleneck): A crossed random-effects decomposition of per-(model, question) Brier@L3 attributes 85.2% of variance to question identity versus 0.5% to model identity (point-estimate ratio 175:1); leave-one-model-out stays in [83.1%, 90.2%]. Question-mix dominates model choice. This extends the finding of Murphy (2026) that question variability accounts for the majority of

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variance in forecaster performance.

Methodological contribution. We propose $\widehat{\text{DRR}}_m^{\text{Res}} = \Delta\text{Resolution}_m - \Delta\text{Resolution}_m | \sqrt{\text{Brier@L0}}$, the residual of the Murphy resolution gain on the prior, as a prior-controlled axis for comparing LLM updating skill. Naive slope is uninformative as a leaderboard metric (τ with Brier@L3 = 0.143, n.s.); $\widehat{\text{DRR}}^{\text{Res}}$ correlates strongly with skill ($\tau = 0.714$, $p < 0.05$) while flipping three model pairs (Table 1). It isolates the updating contribution to skill that Brier@L3 conflates with prior accuracy.

2. Related Work

LLM forecasting. Halawi et al. (2024) introduced the dominant retrieval-augmented pipeline; subsequent work hardened evaluation (Karger et al., 2024; Paleka et al., 2025), explored ensembling (Schoenegger et al., 2024), self-improvement via outcome-driven DPO (Turtel et al., 2025), agentic supervision (Alur et al., 2025), and decomposed evaluation pipelines (Yang et al., 2025). Most of this literature reports *static* accuracy at a single time point.

Calibration under updating. Yuan et al. (2025) is the most direct prior study of how LLM forecasts respond to new information; it concludes that LLMs update conservatively and inconsistently relative to human forecasters. We sharpen that observation by replacing heuristic rational references with a Manifold human-crowd anchor. We also show that the natural follow-on hypothesis, that update slope is a calibration fingerprint, fails as a stand-alone measurement.

Question-level variance. Murphy (2026) reports that question variability accounts for the majority (roughly 62%) of variance in forecaster performance under a mixed-effects analysis on ForecastBench. We replicate the directional finding under a distinct crossed random-effects estimator and at greater magnitude (85.2% versus 0.5%, with the between-model component near the boundary of identifiability; leave-one-model-out bounds the question share to [83.1%, 90.2%]).

3. Method

3.1. The Evidence Ladder

For each resolved binary question q with truth $y_q \in \{0, 1\}$ we construct a four-level ladder (c_0, c_1, c_2, c_3): c_0 empty (L0), c_1 neutral background (L1), c_2 a 2–4 sentence factual evidence summary (L2), c_3 1–2 paragraphs of corroborating facts + analysis (L3). No level states the resolution. Ladders are generated by Opus 4.7 at temperature 0; we report a same-family robustness ablation in Appendix E.

3.2. Elicitation and Slope

Given a model m , question q , and level $\ell \in \{0, \dots, L\}$ ($L=3$), we draw $K=3$ probability forecasts and report $\bar{p}_{m,q,\ell}$. The *truth-aligned* forecast is $a_{m,q,\ell} = \bar{p}_{m,q,\ell}$ if $y_q = 1$ and $1 - \bar{p}_{m,q,\ell}$ otherwise. The *update slope* is the OLS coefficient of $a_{m,q,\ell}$ regressed on ℓ , and $\bar{\beta}_m$ is its per-model mean.

3.3. The Prior Bound

Proposition 3.1 (Slope is bounded by prior miscalibration). *Let $a_{m,q,0}, \dots, a_{m,q,L} \in [0, 1]$ be a monotone non-decreasing truth-aligned trajectory with $a_{m,q,L} \leq 1$. Set $k = \lceil (L+1)/2 \rceil$. Then the OLS slope satisfies*

$$\beta_{m,q} \leq K_L (1 - a_{m,q,0}), \quad K_L = \frac{6k(L-k+1)}{L(L+1)(L+2)}.$$

For $L = 3$ this gives $K_3 = 0.4$. Since $\text{Brier@L0}_{m,q} = (1 - a_{m,q,0})^2$, equivalently $\beta_{m,q} \leq K_L \sqrt{\text{Brier@L0}_{m,q}}$.

Sketch. The OLS slope is $\beta = \sum_{\ell} (\ell - L/2) a_{\ell} / \sum_{\ell} (\ell - L/2)^2$, and the constant term in a_{ℓ} contributes zero to the numerator. Under monotonicity with a_0 fixed, the numerator is maximised by minimising a_{ℓ} where $(\ell - L/2) < 0$ and maximising a_{ℓ} where $(\ell - L/2) > 0$, which yields $a_{\ell} = a_0$ for $\ell < k$ and $a_{\ell} = 1$ for $\ell \geq k$. Direct computation gives $\sum_{\ell} (\ell - L/2) a_{\ell} = (1 - a_0) k (L - k + 1)/2$ and $\sum_{\ell} (\ell - L/2)^2 = L(L+1)(L+2)/12$, yielding the bound. $\square$

We assume monotonicity because each ladder rung is designed to add information; the expected truth-aligned forecast under proper ladder construction is non-decreasing in ℓ . Per-model mean trajectories, which are what enter $\bar{\beta}_m$, are monotone end-to-end for 4 of 7 models; the three that violate do so by a maximum of 0.022 at the L0→L1 step. All observed $\bar{\beta}_m$ values lie below 0.13, consistent with the bound at any reasonable prior, and the structural coupling between slope and prior calibration that the proposition predicts is visible across the panel (Figure 1). Per-(model, question) trajectories on $K=3$ temperature samples are noisier (34% weakly monotone end-to-end, 61% on $\ell \in \{1, 2, 3\}$), with most violations concentrated at the L0→L1 step.

Consequence. If two models differ only in prior calibration, the one with better priors has *less room to move* and therefore a mechanically smaller maximum slope. Cross-model correlations of $\bar{\beta}_m$ with any post-evidence skill metric (Brier@L3, log-loss) are confounded by Brier@L0. Slope-skill rankings cannot be read off without conditioning on the prior; we propose $\widehat{\text{DRR}}_m = \bar{\beta}_m - \bar{\beta}_m | \sqrt{\text{Brier@L0}}$ as the right axis.

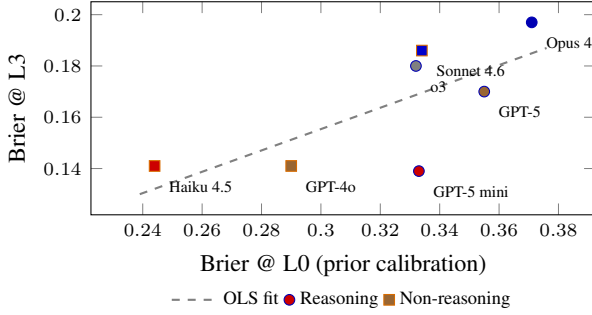


Figure 1. Per-model Brier@L0 vs. Brier@L3, with OLS fit overlaid. Bootstrap-95% CI on Pearson r : [0.251, 0.978] (point estimate 0.716). Prior calibration alone is a strong predictor of post-evidence skill, consistent with Proposition 3.1.

3.4. Manifold Capture Ratios

For each question q with Manifold market consensus $P_{\text{early}}(q)$ (bet-count-weighted mean post-bet probability in the first 20% of the open-trading window) and $P_{\text{late}}(q)$ (last 20%), truth-aligned to y_q , we report two ratios per (model, question):

$$R_{\text{total}} = \frac{a_{m,q,3} - a_{m,q,0}}{P_{\text{late}} - P_{\text{early}}},$$

$$R_{\text{marginal}} = \frac{a_{m,q,3} - a_{m,q,0}}{P_{\text{late}} - a_{m,q,0}}.$$

R_{marginal} removes the concern that R_{total} conflates “the LLM had a better prior than uninformed traders” with “the LLM updates conservatively”; it measures how much of the gap between the LLM’s own prior and the human posterior the ladder closes. Both are computed on questions with $|\text{denominator}| \geq 0.05$.

4. Experimental Setup

Questions. 40 resolved binary questions from Manifold Markets (volume ≥ 1000 , ≥ 25 unique bettors, substantive descriptions, personal/sports markets excluded by regex), resolution dates 2025–2026.

Models. Four reasoning models (GPT-5, o3, Opus 4.7, GPT-5 mini) and three non-reasoning baselines (GPT-4o, Sonnet 4.6, Haiku 4.5).

A contamination check (correlation of per-question mean Brier@L0 with resolution date) is reported in Appendix G; we find no signature consistent with simple memorisation.

5. Results

5.1. Naive slope-vs-skill is unidentifiable; prior alone is not

The simple slope-vs-Brier@L3 correlation in our panel is $r = -0.054$ with bootstrap 95% CI $[-0.848, 0.781]$,

Model	R?	Br@L3	$\bar{\beta}_m$	$\text{DRR}_m^{\text{Res}}$
GPT-5 mini	Y	0.139	0.080	+0.035
GPT-4o	N	0.141	0.066	+0.030
GPT-5	Y	0.170	0.069	+0.012
o3	Y	0.180	0.053	+0.003
Haiku 4.5	N	0.141	0.049	-0.021
Opus 4.7	Y	0.197	0.058	-0.025
Sonnet 4.6	N	0.186	0.079	-0.034

Table 1. Per-model leaderboard, sorted by $\text{DRR}_m^{\text{Res}}$. Raw slope $\bar{\beta}_m$ rank correlates with Brier@L3 only at Kendall $\tau = 0.143$ (n.s.); $\text{DRR}_m^{\text{Res}}$ at $\tau = 0.714$ ($p < 0.05$) while flipping three pairs against raw Brier@L3.

unidentifiable from 7 models. Per-model Brier@L0, in contrast, predicts Brier@L3 with bootstrap CI bounded away from zero ($r = 0.716$, CI [0.251, 0.978]; Figure 1). The structural reason is Proposition 3.1: slope is mechanically constrained by $\sqrt{\text{Brier@L0}}$, with empirical OLS slope on $\sqrt{\text{Brier@L0}}$ of 0.131.

Hierarchical confirmatory test. The full slope $\beta_{m,q}$ shares the $a_{m,q,3}$ term with Brier@L3 mechanically, so a regression of Brier@L3 on $\beta_{m,q}$ would partly recover that identity. We therefore use the $L0 \rightarrow L2$ sub-slope $\beta_{m,q}^{02} = (a_{m,q,2} - a_{m,q,0})/2$, which is a function of a_0, a_1, a_2 only and is out-of-sample for a_3 , in a crossed mixed-effects model $\text{Brier@L3}_{m,q} = \alpha + \gamma_1 \beta_{m,q}^{02} + \gamma_2 \text{Brier@L0}_{m,q} + u_m + u_q + \varepsilon$ on $n=268$ cells. Without prior control, $\hat{\gamma}_1 = -0.531$ (SE 0.112, CI $[-0.752, -0.311]$); after adding Brier@L0, $\hat{\gamma}_1 = -1.781$ (SE 0.086, CI $[-1.949, -1.614]$). The coefficient roughly triples in magnitude after the prior is controlled, which is the prior-confound signature Proposition 3.1 predicts: sub-slope is positively coupled with prior miscalibration (low-prior models have more room to update), so the marginal coefficient understates the conditional updating-skill effect.

5.2. Dose-response residual ranks models on updating skill

Table 1 reports $\text{DRR}_m^{\text{Res}}$ alongside raw slope. With 7 models the implied ordering is exploratory; the point of Table 1 is the methodology, not a confirmatory ranking. $\text{DRR}_m^{\text{Res}}$ is the right axis to report alongside (or instead of) raw slope when comparing LLM forecasters on dose-response benchmarks. Note that the top and bottom of the $\text{DRR}_m^{\text{Res}}$ ranking both include reasoning and non-reasoning models, further arguing against treating naive slope as a paradigm-level fingerprint.

5.3. LLMs capture ~40% of the human-crowd update

We anchor LLM updates against the Manifold human-crowd update on the same questions. The aggregate panel mean

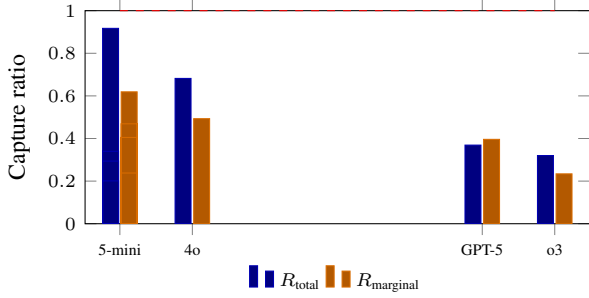


Figure 2. Per-model median Manifold capture ratios. Dashed red line: ratio of 1.0 (LLM update matches human-crowd update). The two metrics agree directionally; the apples-to-apples concern that R_{total} conflates better LLM priors with conservative updating is not borne out.

risers from 0.521 at L0 to 0.687 at L3. Both ratios agree: R_{total} median 0.397 (CI [0.252, 0.572], 245 pairs) and R_{marginal} median 0.396 (CI [0.256, 0.490], 239 pairs). The interpretations differ: R_{total} measures the fraction of the human-crowd update the LLM reproduces, while R_{marginal} measures the fraction of the gap between the LLM’s own prior and the human posterior that the ladder closes. **Both come in at $\sim 39.6\%$, well below 1.0.** The agreement removes the concern that one ratio is inflated by LLM prior knowledge that uninformed traders lack; the LLM does not match the human-crowd response on evidence on either metric.

Per-model capture ratios (Figure 2) range from 0.20 (Opus 4.7) to 0.92 (GPT-5 mini) on R_{total} , and 0.23 to 0.62 on R_{marginal} . Only GPT-5 mini exceeds 0.6 on the marginal ratio, the cleaner conservatism measure; six of seven models cluster below 0.5.

5.4. Question identity explains 85.2% of variance

A crossed random-effects model $\text{Brier@L3}_{m,q} = \mu + u_q + v_m + \varepsilon_{m,q}$ fit on the 7×40 cells yields $\hat{\sigma}_q^2 = 0.0544$, $\hat{\sigma}_m^2 = 0.0003$, $\hat{\sigma}_\varepsilon^2 = 0.0091$, for variance shares of 85.2% (question) and 0.5% (model). The between-model variance component is estimable but at the small end of the parameter space; the point estimate ratio is 175:1. Leave-one-model-out re-estimation bounds the question share to [83.1%, 90.2%]. We take this as evidence that benchmark design, not model selection, is the principal lever for the field’s measurable progress on forecasting.

A Murphy decomposition (reliability – resolution + uncertainty, five equal-width bins) shows that L0 to L3 increases mean resolution by 0.110 versus reducing reliability by 0.046 ($\Delta_{\text{res}} > |\Delta_{\text{rel}}|$ for six of seven models): the ladder’s effect on skill is primarily through discrimination, not confidence calibration. We use this to motivate $\text{DRR}_m^{\text{Res}}$ above; the cross-model correlations between L0 and L3 Murphy

components are exploratory at $n=7$ and we make no confirmatory claim from them.

6. Discussion

What an honest dose-response evaluation looks like. Proposition 3.1 forces a redesign: plot $\text{DRR}_m^{\text{Res}}$ rather than $\hat{\beta}_m$, anchor “how much did the model update” to a measured human-crowd reference, and stratify Brier by question.

The variance-bottleneck consequence. With 85.2% of variance carried by question identity and the between-model component at the small end of identifiability, percentage-point differences between models on any AI-forecasting leaderboard drawn from a different question pool are statistically indistinguishable from question-mix noise. The implication is not that all LLM forecasters are equivalent in some deep sense, but that the *instrument* (a benchmark with a fixed question pool) has lower discriminating power than the leaderboard reports imply. Benchmarks should report question-stratified Brier rather than aggregate Brier, and treat leaderboard movement inside the σ_{question} band as null until shown otherwise.

Limitations. First, our panel has $n=7$ models and the cross-model correlations therefore have wide bootstrap CIs; we report point estimates as exploratory and rely on structural arguments and the crossed random-effects fit for confirmatory claims. Second, ladders were generated by Opus 4.7, which is in the elicitation panel; we provide a same-family ablation in Appendix E. Third, the question distribution is Manifold-Markets-specific; the prior-confound result is structural and should replicate, but the specific numerical magnitudes are sample-dependent.

7. Conclusion

Slope is a mechanical function of headroom, and headroom is a function of prior calibration. The cleaner anchor for “how much did the LLM update” is a measured human-crowd update on the same questions; against that anchor, LLMs close roughly 39.6% of the gap between their prior and the human posterior (CI [0.256, 0.490]). Between-model variance in raw Brier@L3 is dwarfed by question-mix variance; once prior calibration is controlled, the prior-controlled updating-skill component ($\text{DRR}_m^{\text{Res}}$) does discriminate models meaningfully, though within a narrow absolute range. Dose-response evaluation is useful, but only with prior control, human anchors, and question stratification.

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A. Reproducibility

Code and data will be released upon publication.

B. Ladder Construction Protocol

Each ladder is generated at temperature 0 by Opus 4.7 with the prompt reproduced verbatim in Appendix C. We audit a random 10% sample of ladders for resolution leakage, defined as a factual statement that uniquely entails y_q . Ladders failing the audit are reconstructed.

C. Elicitation Prompt

You are a careful probabilistic forecaster.

QUESTION: {title}
 RESOLUTION CRITERIA: {criteria}
 CONTEXT:
 {context}

Give your forecast as a single probability that the question resolves YES.

Reply with exactly one line in this format and nothing else:

PROBABILITY: 0.XX

D. Manifold Anchor: Per-Model Capture Ratios

Model	R?	Median R_{total}	Median R_{marginal}
GPT-5 mini	Y	0.917	0.619
GPT-4o	N	0.682	0.493
Opus 4.7	Y	0.201	0.238
Sonnet 4.6	N	0.294	0.405
Haiku 4.5	N	0.340	0.469
GPT-5	Y	0.369	0.396
o3	Y	0.320	0.234

Table 2. Per-model median capture ratios (total and marginal). Capture ratios closer to 1 indicate that the LLM internalises a larger fraction of the human-crowd update on the same evidence.

E. Ladder Same-Family Ablation

To test whether Opus 4.7’s per-question slope is artificially inflated when reading Opus-authored ladders, we regenerate 10 ladders with GPT-5 (a non-Anthropic-family builder) and re-elicite Opus 4.7 on the new ladders. Per-question Opus slopes are then compared pair-wise.

Opus’s mean per-question slope was 0.0481 on Opus-authored ladders versus 0.0540 on GPT-5-authored ladders (mean difference -0.0059 ; paired t -test $p=0.76$, Wilcoxon signed-rank $p=0.846$). The point estimate is in the opposite direction from the contamination prediction (Opus is marginally *flatter* on its own ladders, not steeper) and is statistically indistinguishable from zero. We take this as bounded evidence against same-family contamination at the ladder-construction stage; a larger ablation with multiple non-panel builders would be stronger.

F. Random-Effects Decomposition

We fit a single crossed random-effects model using `statsmodels.mixedlm` with `vc_formula` providing a crossed Model component on top of a Question random intercept. The fit yields $\hat{\sigma}_q^2 = 0.0544$, $\hat{\sigma}_m^2 = 0.0003$, $\hat{\sigma}_\varepsilon^2 = 0.0091$, giving the variance shares quoted in §5.4. Leave-one-model-out re-estimation yields question-shares ranging from 83.1% to 90.2% across the 7 holdouts.

G. Contamination Check Details

Per-question Pearson correlation of mean Brier@L0 with resolution date was -0.362 ($p = 0.0219$). On a median-resolution-date split, the earlier half ($n = 20$) had mean Brier@L0 = 0.403 versus the later half ($n = 20$) = 0.242. The simplest memorisation prediction (recent questions worse-calibrated) is not observed; we make no positive claim about contamination beyond the absence of this specific signature.